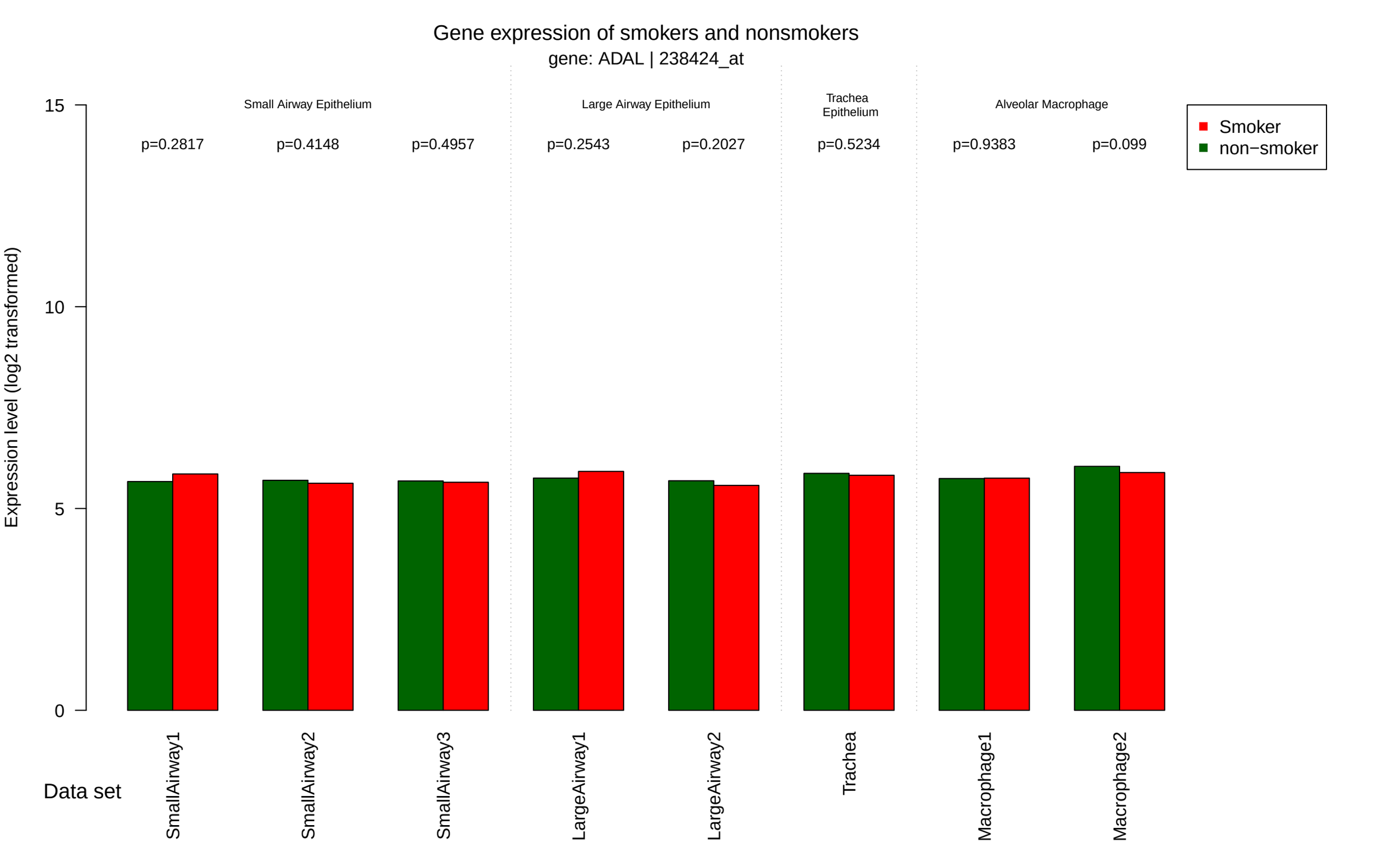
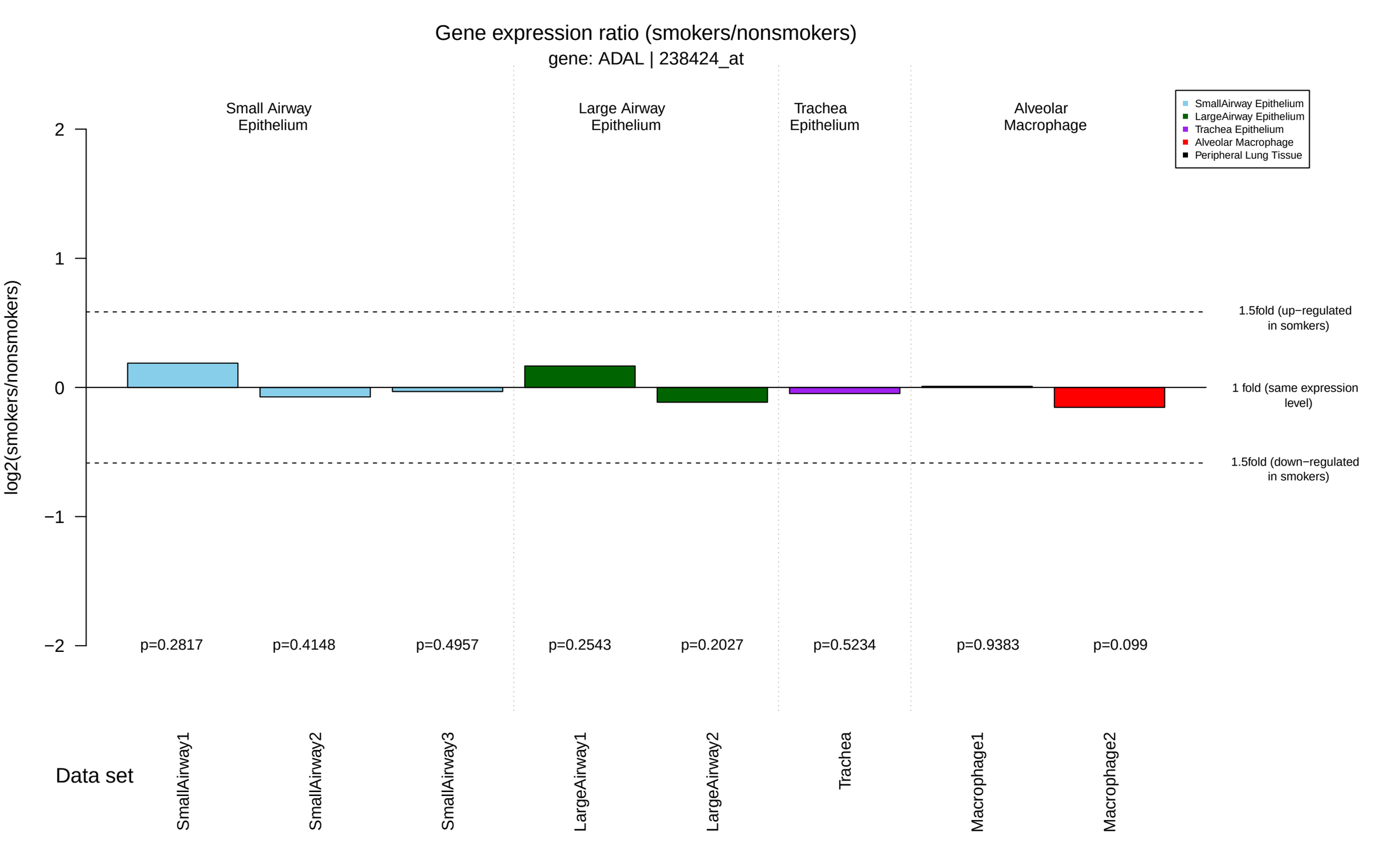
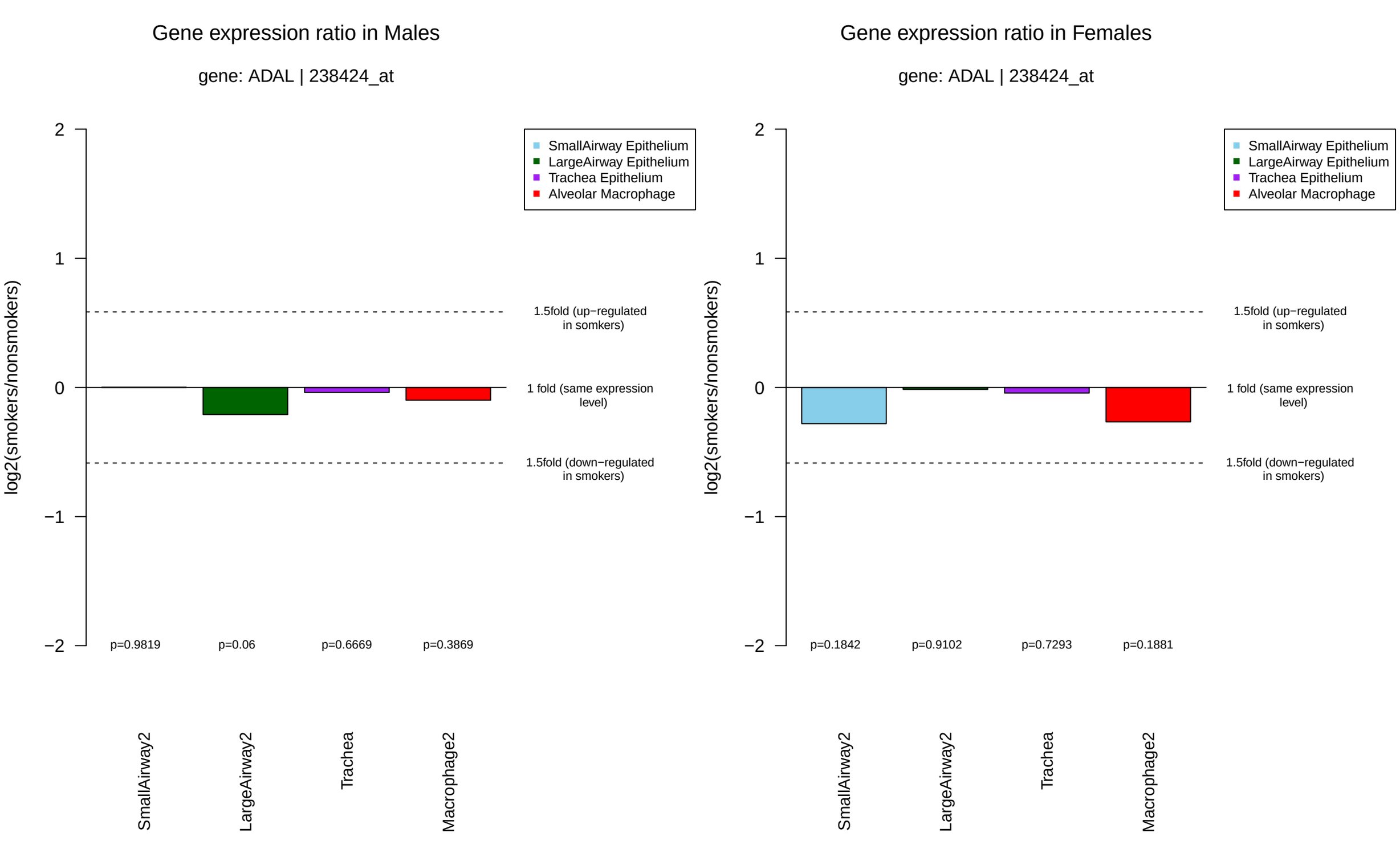
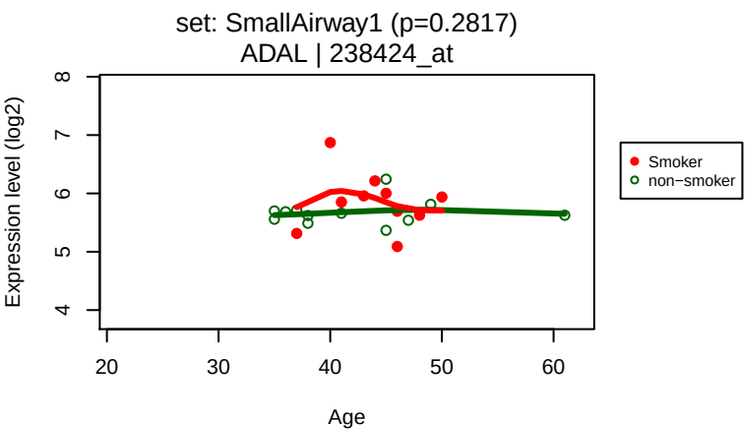
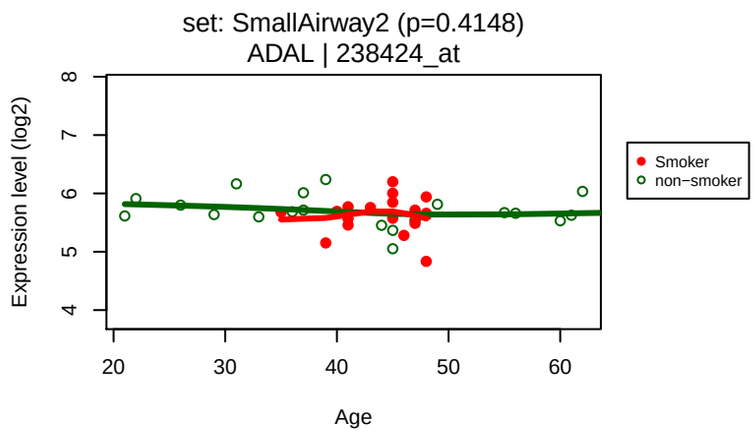
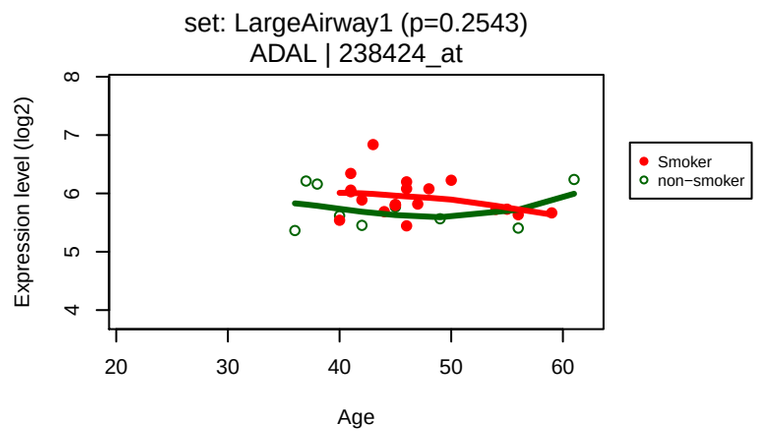

Gene expression of smokers and nonsmokers
gene: ADAL | 238424_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2817
p=0.4148
p=0.4957
p=0.2543
p=0.2027
p=0.5234
p=0.9383
p=0.099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

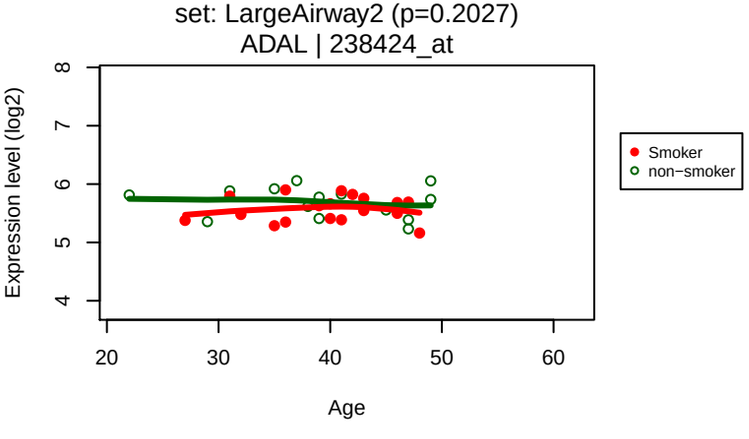

Gene expression ratio (smokers/nonsmokers)
gene: ADAL | 238424_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2817
p=0.4148
p=0.4957
p=0.2543
p=0.2027
p=0.5234
p=0.9383
p=0.099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

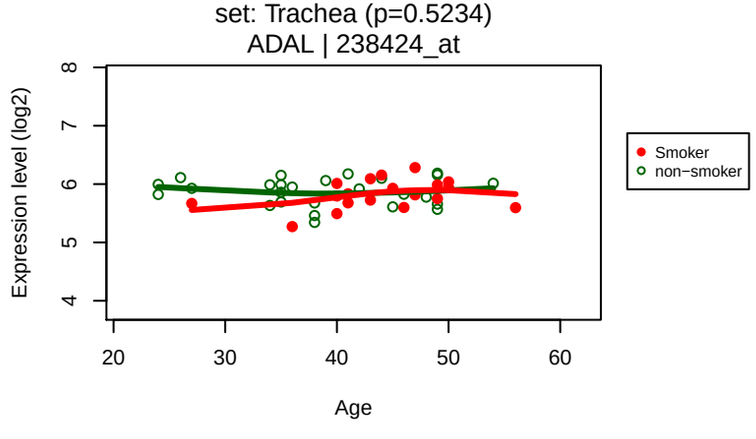

Gene expression ratio in Males
gene: ADAL | 238424_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9819
p=0.06
p=0.6669
p=0.3869
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAL | 238424_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1842
p=0.9102
p=0.7293
p=0.1881
SmallAirway2
LargeAirway2
Trachea
Macrophage2

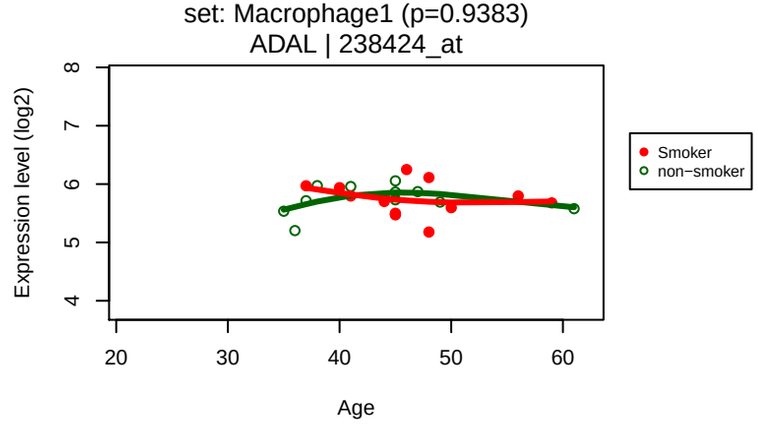

set: SmallAirway1 (p=0.2817)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker

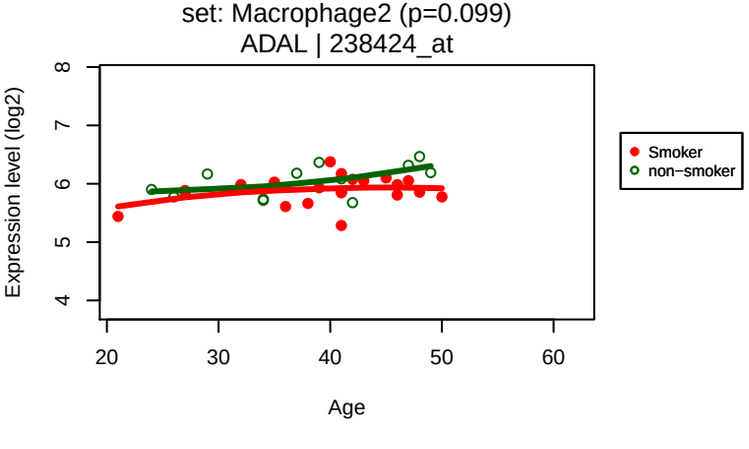

set: SmallAirway2 (p=0.4148)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker

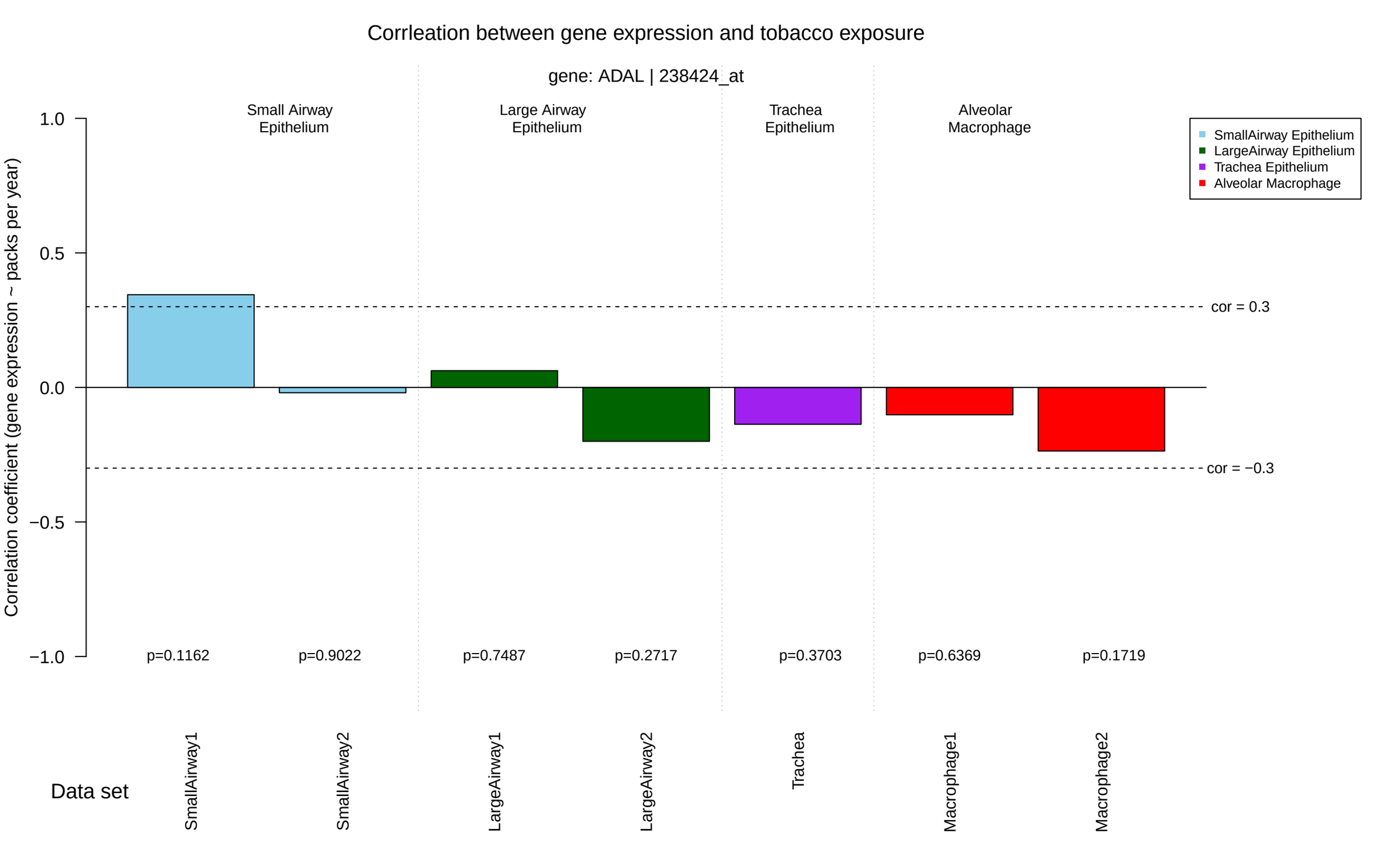

set: LargeAirway1 (p=0.2543)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2027)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5234)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9383)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.099)
ADAL | 238424_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAL | 238424_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1162
p=0.9022
p=0.7487
p=0.2717
p=0.3703
p=0.6369
p=0.1719
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2